Activation Plateaus Meeting Slides

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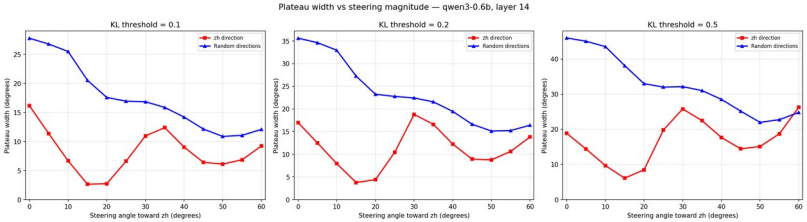
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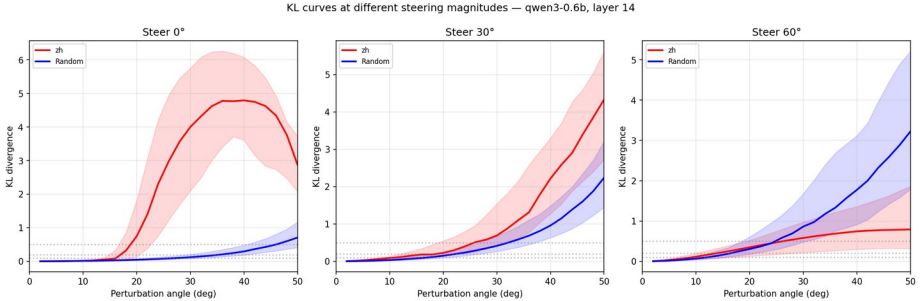
Last time: multi-contrastive steering, adapted MELBO, auto-interp, discussed research directions.

Today:

1. Plateau-ness degrades away from real activations
2. Cross-group feature composition
3. Two dead ends (Jacobian directions, sparse models)
4. Quantitative model for plateau-breaking angles
5. ChessGPT: three-regime validation
6. Subspace scaling — 25 features
7. Feature refinement + progressive discovery (~140 features)
8. Subspace scaling — 135 features with variable thresholds
9. Feature reconstruction
10. Discussion / next steps

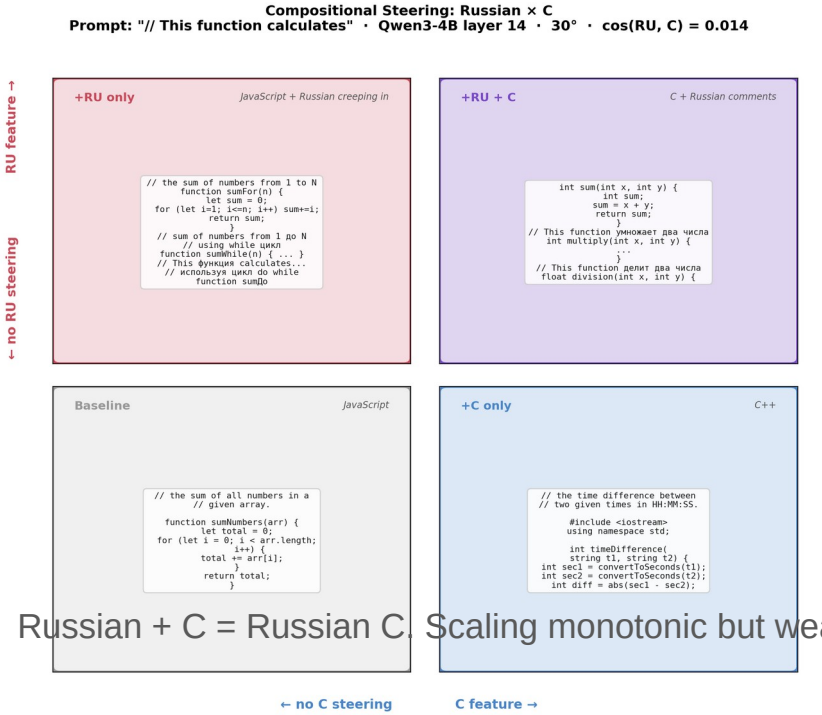
Plateau-ness degrades away from real activations

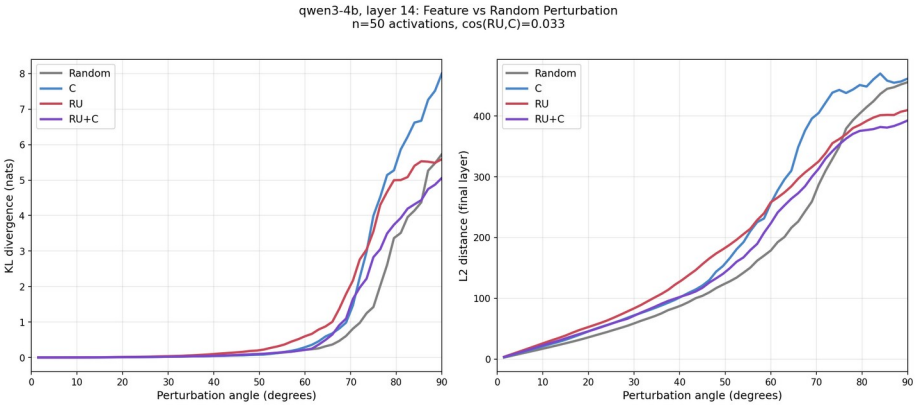




Steer at increasing magnitudes, measure plateau width. Plateau shrinks. Addresses direction 1.5.

Cross-group feature composition

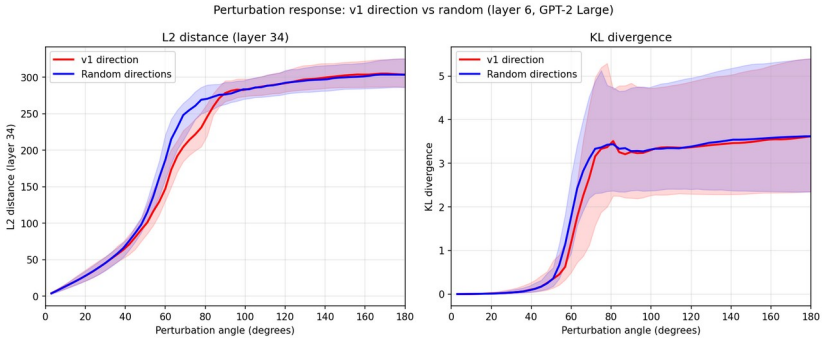




Russian + C = Russian C. Scaling monotonic but weak. Methodological note: must include dominant language.

Two dead ends

Jacobian-norm directions aren't features



Sparse models lack plateaus

Pivoted to ChessGPT.

Quantitative model for plateau-breaking

Key equation: $\theta_{break}(d) = \arcsin(\sin(\theta_1) / \alpha(d))$

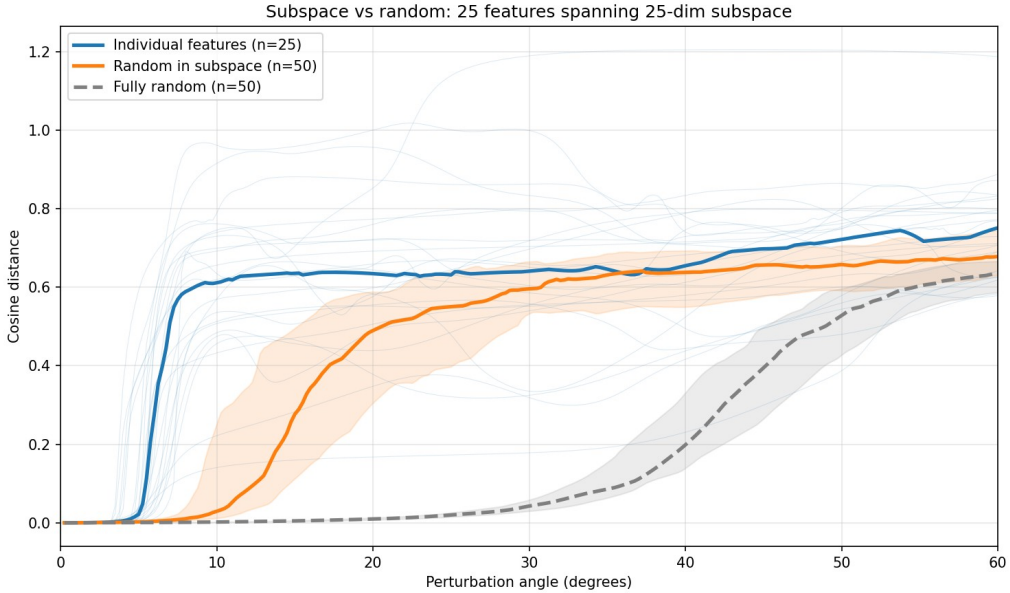
where $\alpha(d) = \max_i |\langle d, f_i \rangle|$ is the max alignment with any feature.

Three predictions:

1. **Single feature:** $\theta_{break} = \theta_1 \approx 5.5°$
2. **Random in k-dim subspace:** $\theta_{sub} = \arcsin(\sin(\theta_1) \cdot \sqrt{(k / 2 \ln k)}) \approx 11°$ for $k=25$
3. **Fully random:** $\theta_{rand} = \arcsin(\sin(\theta_1) \cdot \sqrt{(n / 2 \ln m)}) \approx 36°$ for $m \approx 1000$

Brief — details in the shared write-up.

ChessGPT: three-regime validation



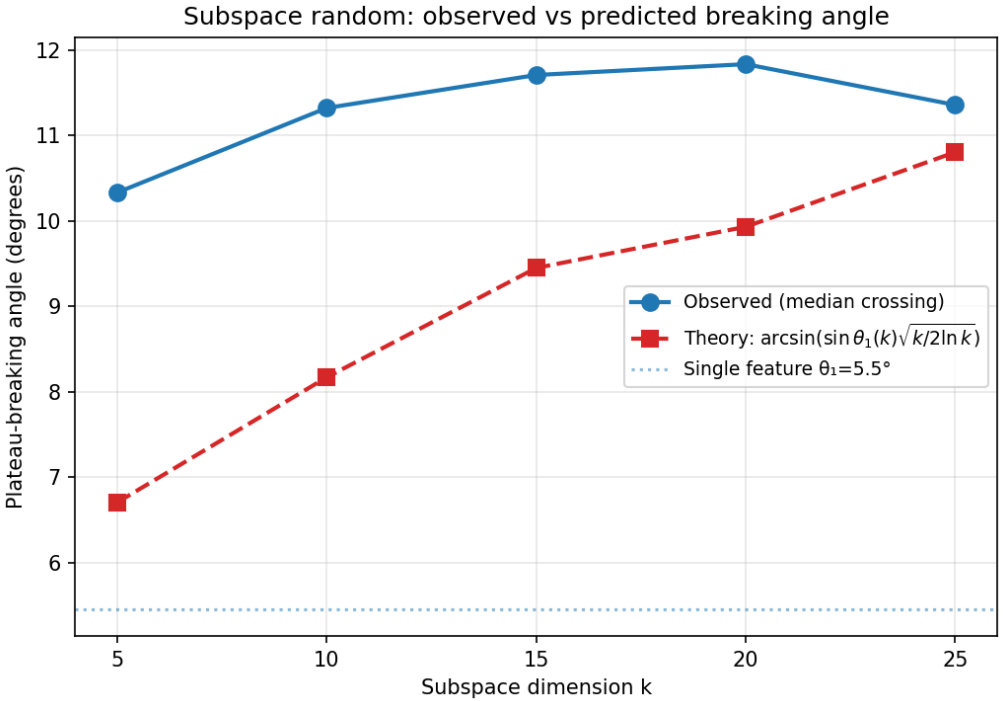
Observed vs predicted:

Features: ~5.5°

Subspace random: ~12.5° (pred: 11°)

Fully random: ~36.3° (pred: 36°, m≈1000)

Subspace scaling — 25 features

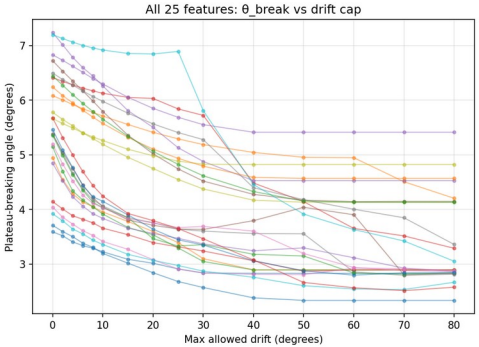


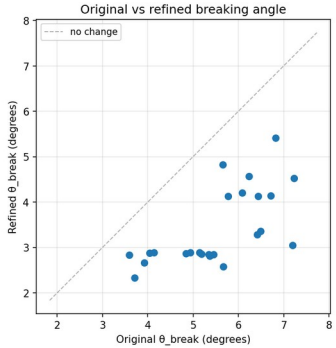
Directionally correct, underpredicts by 20–30%.

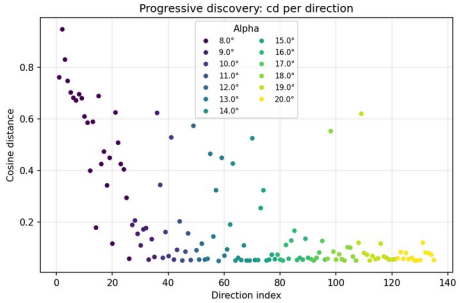
Two confounders:

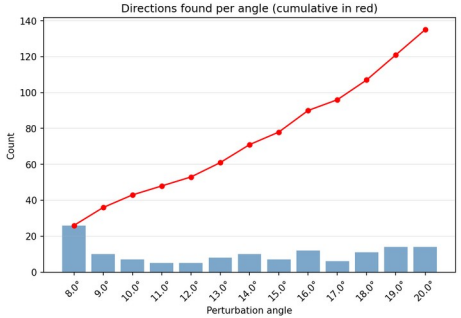
1. Dirty features (rotated from true features)
2. Variable θ_1 per feature (model assumes uniform)

Feature refinement + progressive discovery





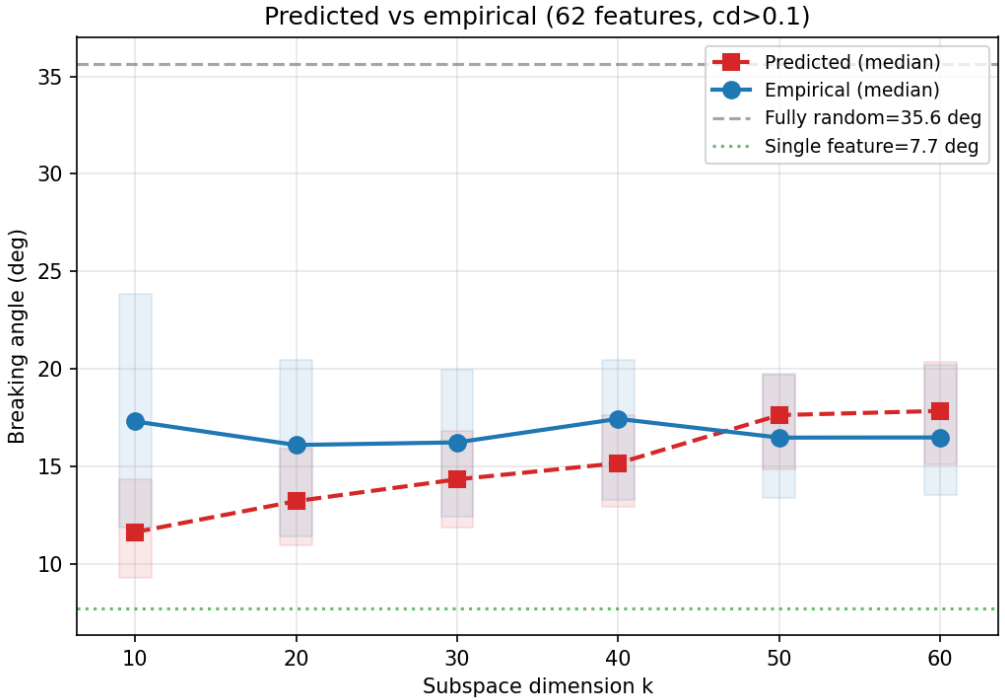




Local optimization cleans features. θ_1 range: 2.2–5.5°.

Progressive MELBO: ~140 features in 512-d space.

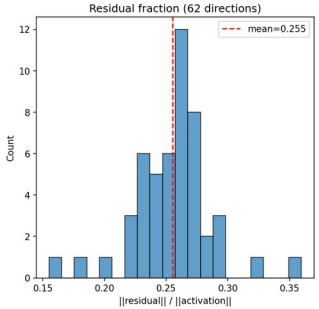
Subspace scaling — 135 features, variable

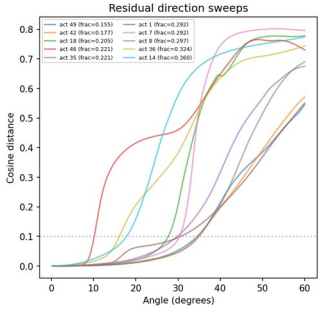


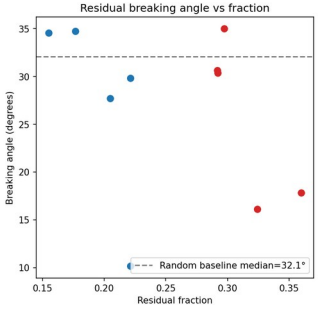
Predicted trend increases, empirical is flat, error bars overlap.

Current status.

Feature reconstruction







62 directions explain 93.4% variance. Top 3 explain 83%.

Residuals have random-like plateau angles.

d1 correlates r > 0.96 with move count.

Discussion / next steps

Reference research directions document.

What's prioritized, what's open for discussion.

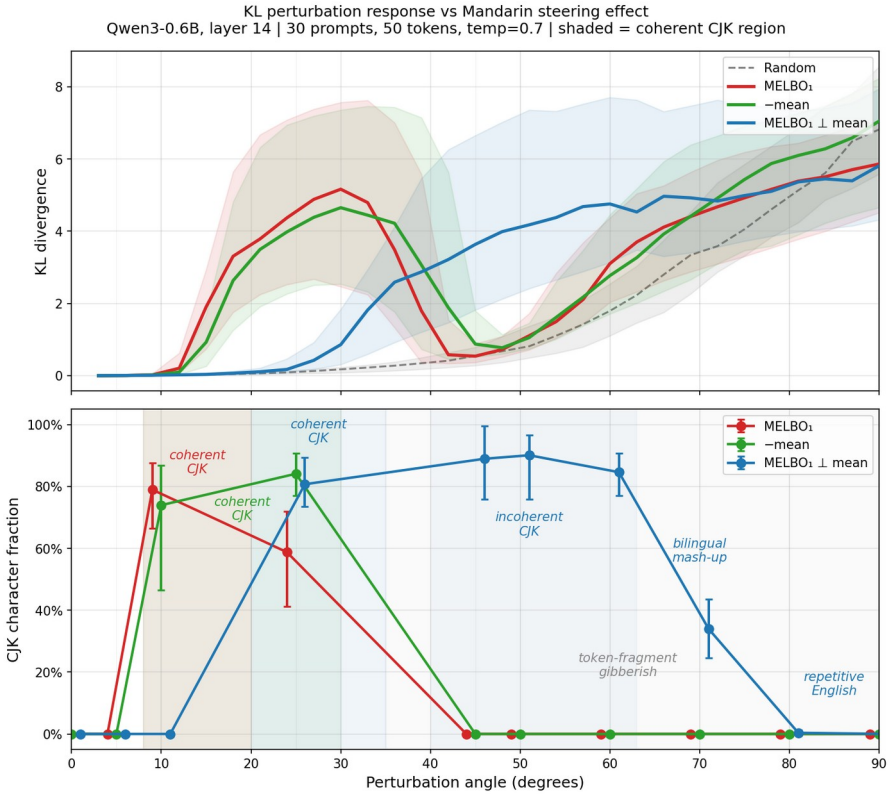
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Last time: Feature directions (PC1/MELBO$_1$) break plateaus while the nearly-identical mean direction does not

Today: Things I've done that I'd like to cover:

1. Solution to the above puzzle
    a. Not so exciting; I was steering the mean in the wrong direction lol
2. Steering lines up nicely with perturbation sweep
3. Multi-contrastive steering: can we estimate feature vector rather than difference-of-means?
4. Adapted MELBO: selecting for feature-like response profiles

Steering and Perturbation Line Up



- Top: perturbation sweep KL-div (sorry)
- Bottom: steering effectiveness as fraction of CJK characters starting from English prompt
- Component of MELBO_1 that's orthogonal to mean also steers, but in a different regime
- Component of mean orthogonal to MELBO_1 does nothing

Multi-contrastive steering/feature isolation

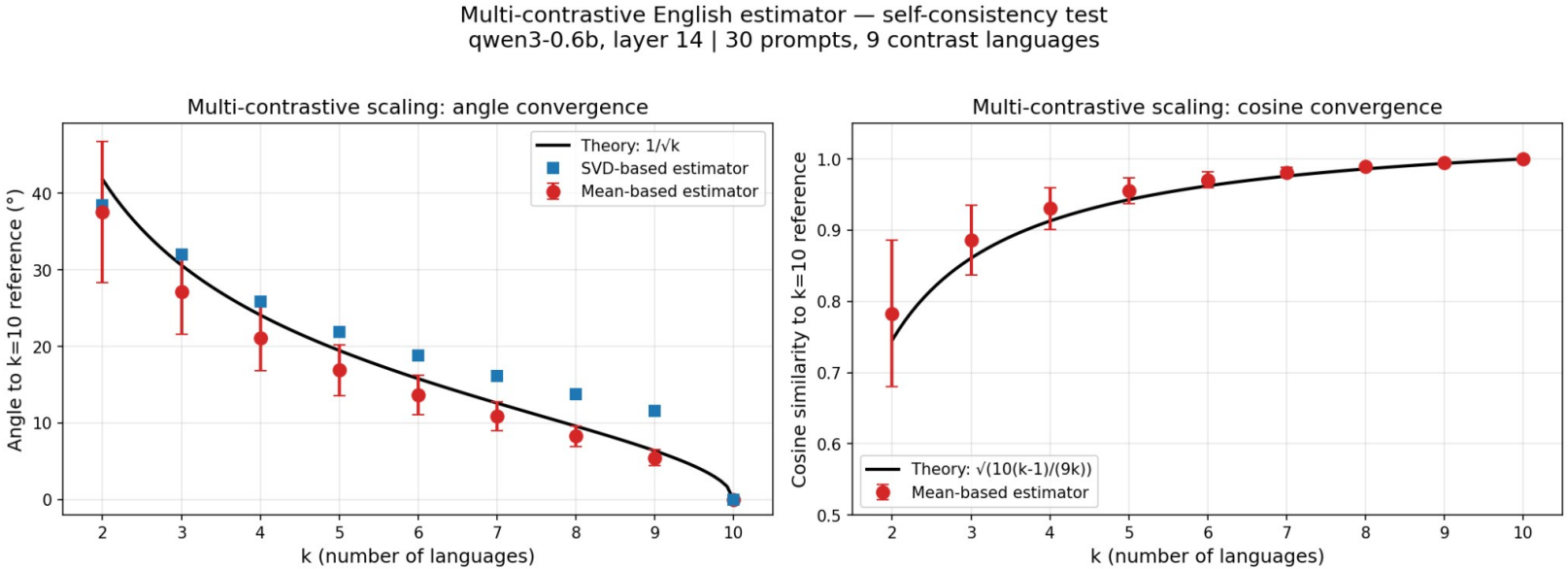
Diff-of-means gives you vector between two concepts, not the concepts' vectors d_EM = mean(EN) - mean(ZH) \approx f_EN - f_ZH

Idea: If we contrast English against many languages, the English component is shared across all differences while the other languages point in other directions and cancel out

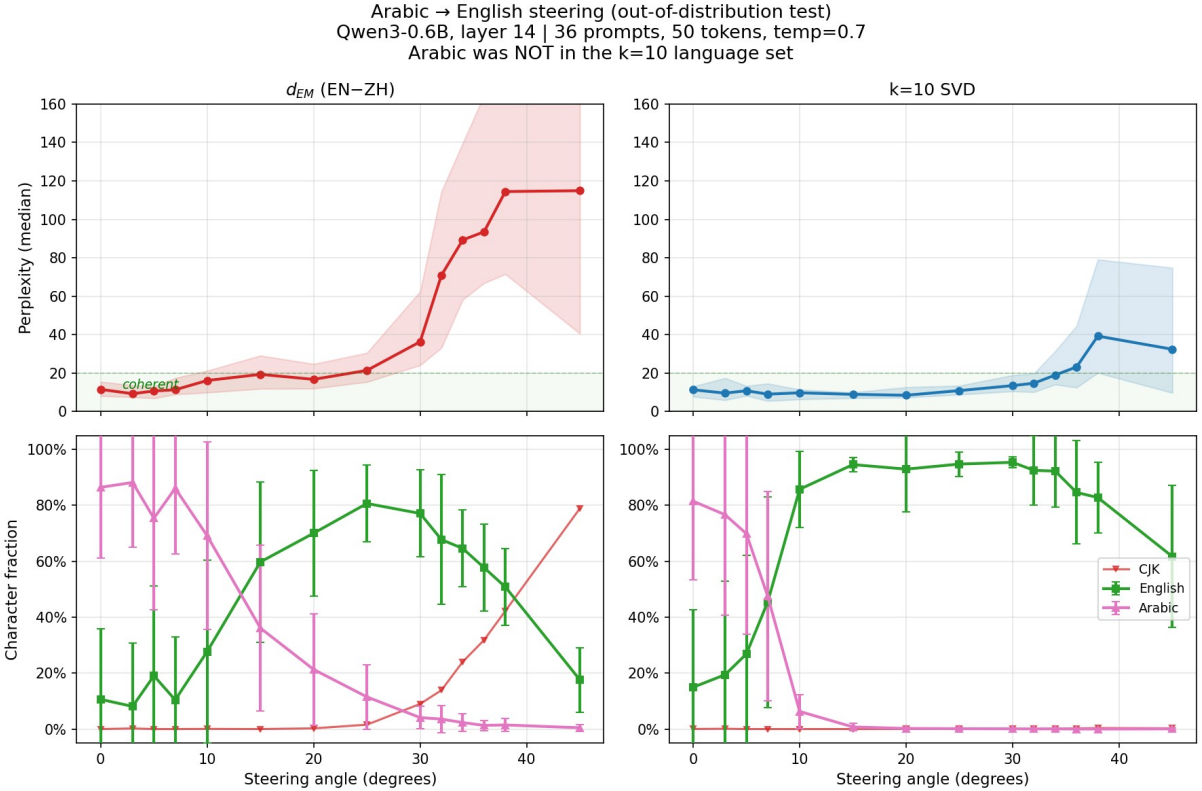
Stack them, SVD -> Isolate f_EN

Validation: convergence

Predict convergence as 1/sqrt(k); effectively independent measurements if features orthogonal

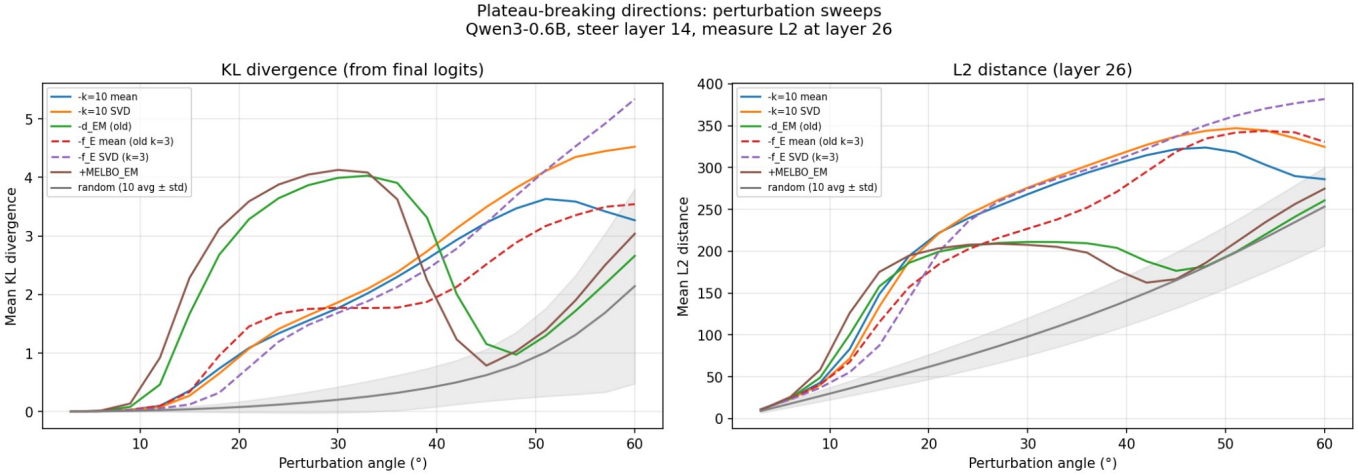


Validation: generalizes to held-out language



- Arabic not in k=10
- Perplexity as proxy for coherence
- Different alphabet -> easy to measure steering effect
- Better than diff-of-means
- Claim: We have English direction

Perturbation sweeps look… smoother?

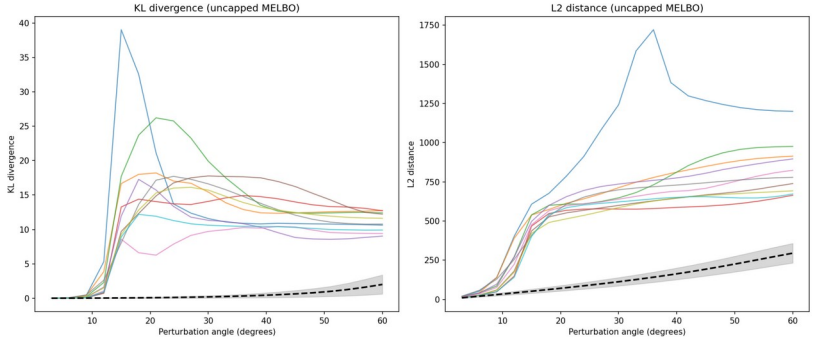


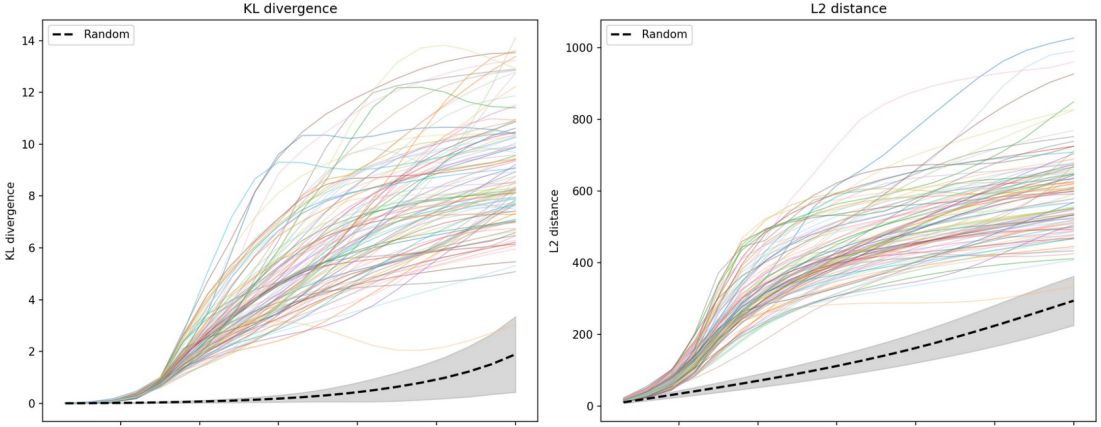
Pretty linear for quite a range after plateau

Do we want to optimize for this profile?

Note: much less notable in L2?

MELBO on diverse dataset
Top: max kl-div; Bottom: profile-matched kl-div





Problem: MELBO directions break the model

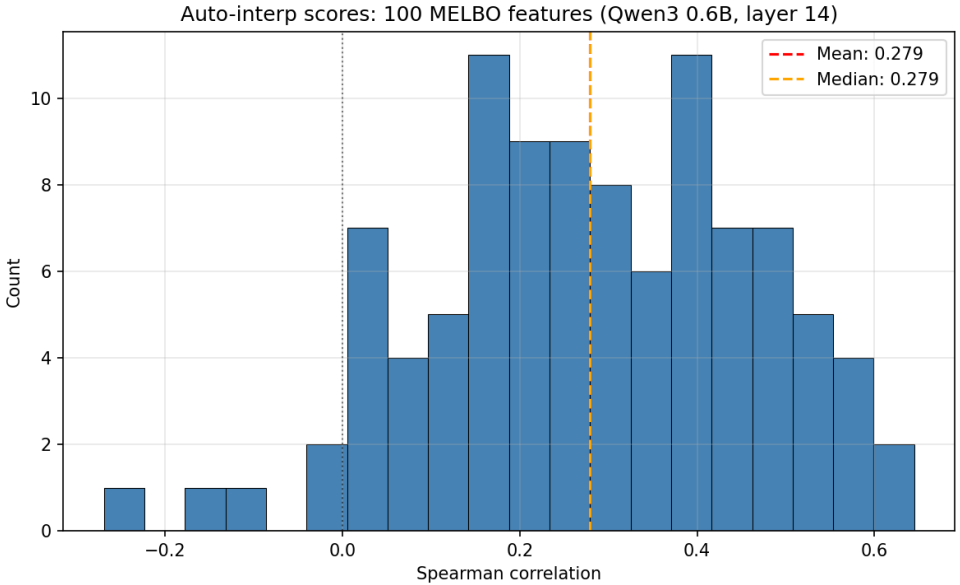
Idea: Instead of direction of max perturbation, find direction that breaks off to ~1 KL-div around ~15 degrees (these are hyperparameters)

Quite a few interpretable! Steered with top ~10 and found:

1. numbers/math-y
2. Verbosity?
3. Economics
4. Academic Publication?

Auto-interp as per Anthropic

Correlate LLM-explainer with LLM-simulator of top-prompts activating directions



Manual check on top few looks decent

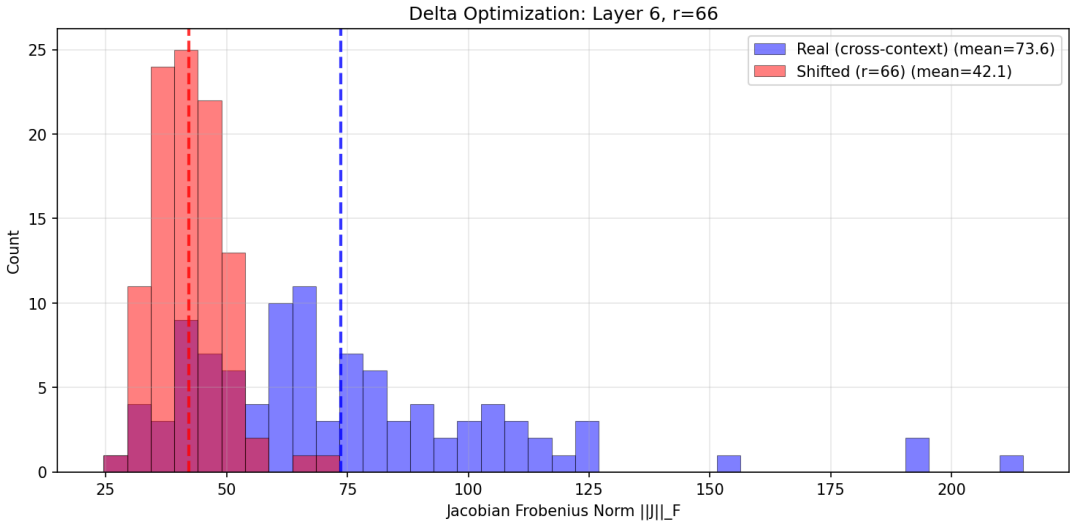
Smaller numbers than Anthropic

Probably too little context: 5 tokens vs 128 their experiment

Questions/Directions

1. How to do interpretability at scale? Steering and looking at results isn't fun for 100 directions… Delegate to LLM? Something else?
2. Tried to generalize multi-contrastive steering to animals but did not work :-)
3. MELBO constrained to span of differences of animals/active dataset dogs also did not work
4. Is this direction interesting? Or should we stick to unconstrained, since we can apparently find features?
5. Keen on the 'geometry of plateau' direction? Feature radius vs random radius, I mean

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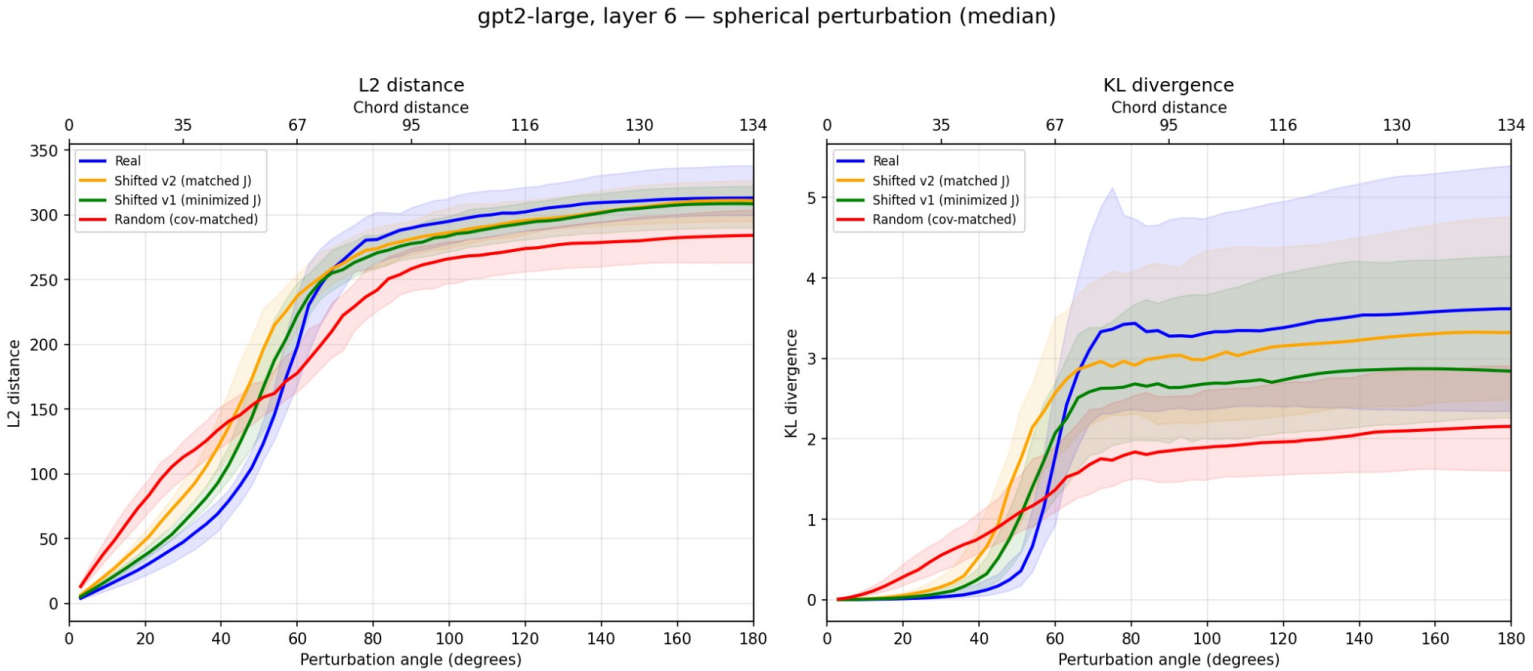
Last time: We can optimize for vector taking us from set of real activations to set of activations with small Jacobian norm!

Up next: A few threads proposed:

1. Do shifted activations have plateaus?
2. Find steering vector that 'works'. Do resulting activations have plateaus? Follow-up: can we optimize them/the steering vector so that they do?
3. How does steering compare plateau-wise to perturbing along random direction?

Do shifted activations have plateaus?

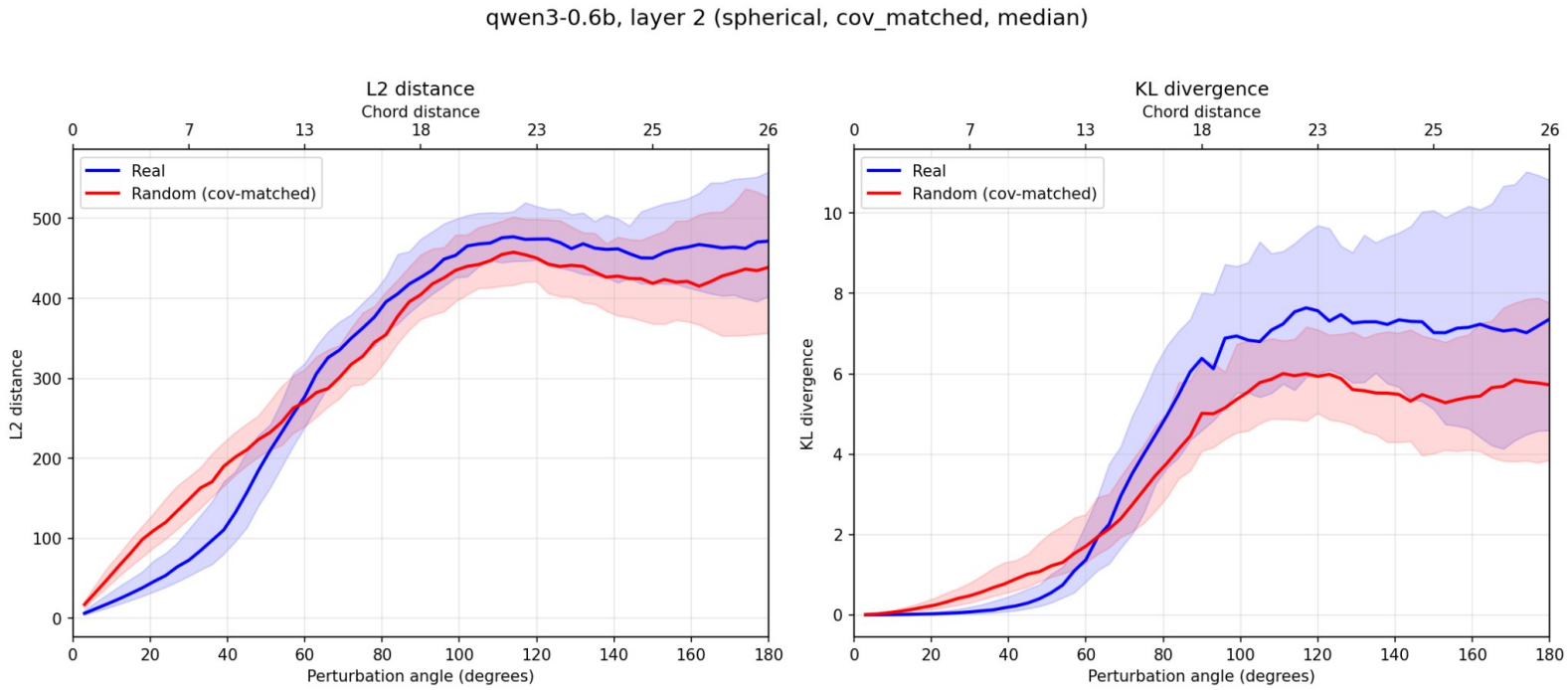
Yes! But smaller than real activations. Matching Jacobian norm not enough; minimizing it is better



Intermezzo: switched to Qwen3 0.6B

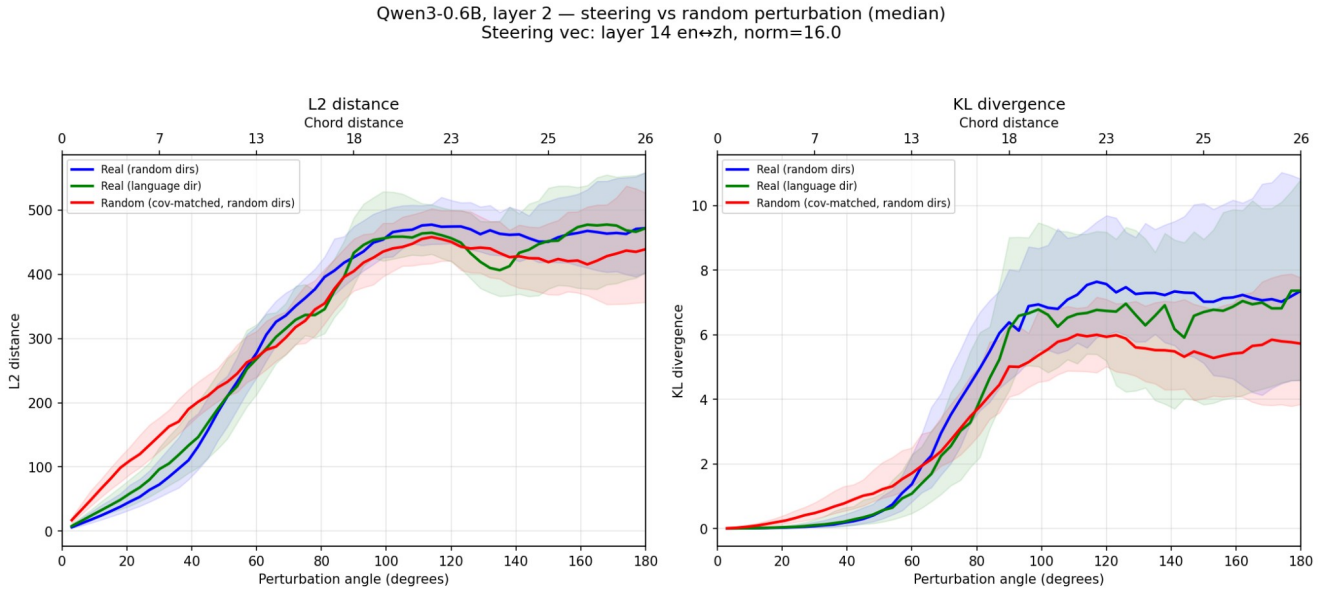
Sanity check: Does it exhibit plateaus?

Aside: Previous mentees of Stefan found no plateaus in Gemma (https://arxiv.org/abs/2409.17113). What's up with that?



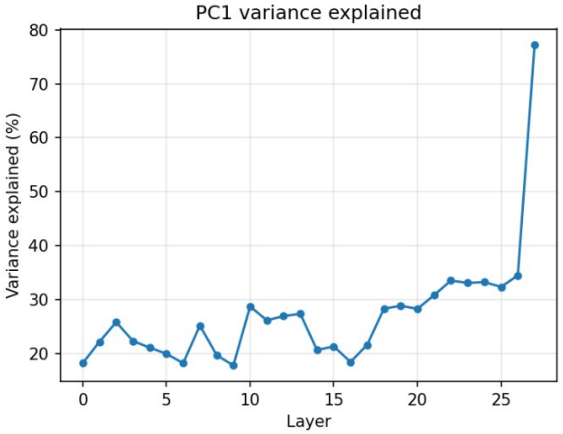
How does steering compare plateau-wise to perturbing along random direction?

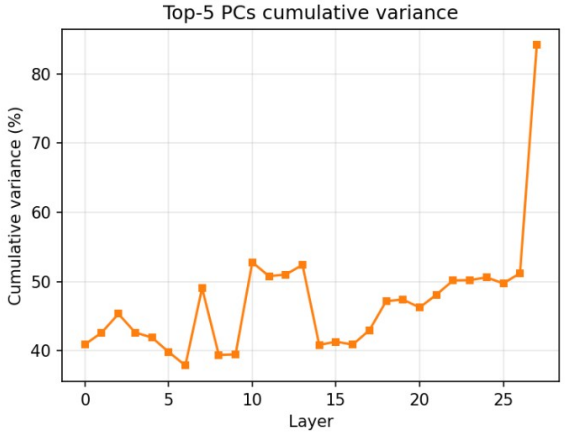
Steering direction: difference-of-means English, Mandarin prompts (generated by Claude, I don't speak Mandarin)
Perturb along steering direction; measure impact on later layer… Looks just like random??
Hypothesis: steering vector is not clean feature, but superposition of many features -> random-like
Potential confounder: perturbing at layer 2, extracted vector at layer 14 (but holds also with both at layer 14; see later slide)

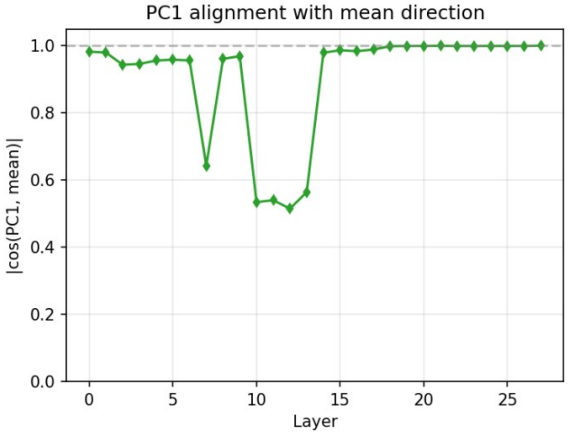


Can we decompose the matrix of differences?

Performed SVD on the matrix of differences at each layer (so stacked vectors of activation differences), checked how much variance is explained by PCs







Dips in mid layers appear to be due to token-length mismatches in some contrastive pairs; removing outliers gets rid of them

At most layers, PC1 extremely aligned with mean; but vector is multidimensional: top PC explains only ~30% of variance

Can we optimize the decomposition of the language steering vector?

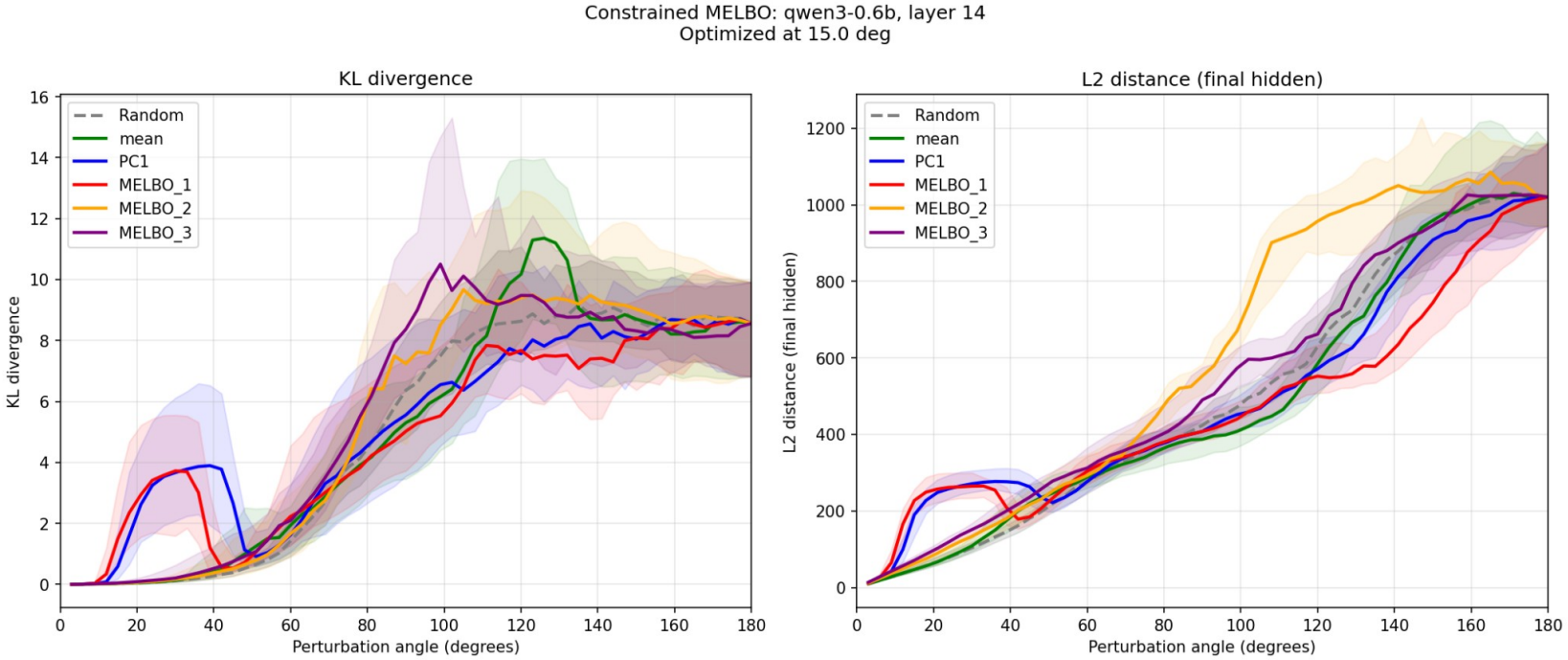
Goal: Find the direction *within the span of the difference matrix D* that maximizes KL divergence when used as a spherical perturbation.

Why? This asks: "which component of the language steering vector causes the most behavioral change?"

Note that this is kind of like constrained MELBO.

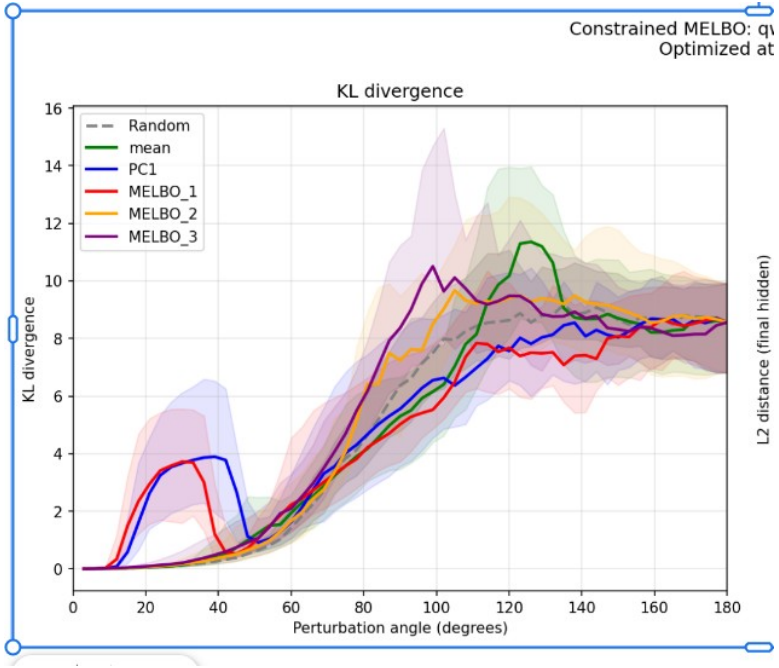
Fix magnitude of perturbation at 15 angles: guess; still in plateau for random direction, but perhaps not for feature direction

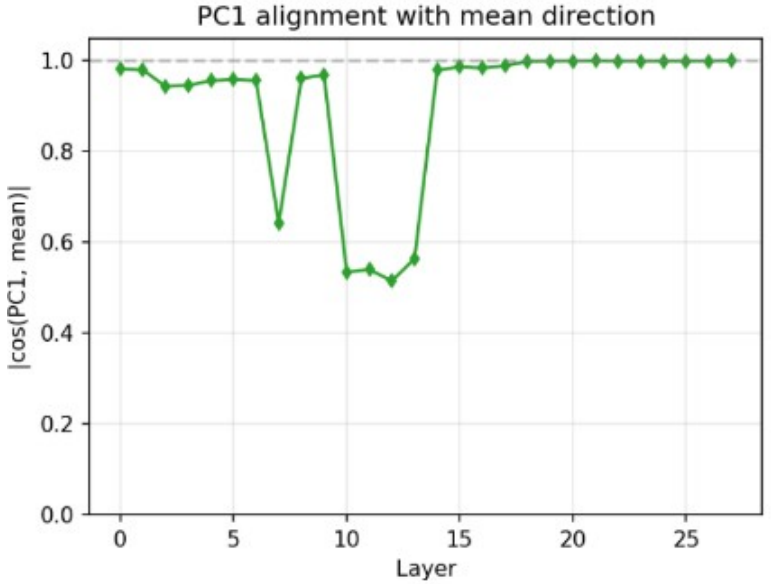
We can, and it breaks the plateau!



So does PC1 - very similar behavior

Is activation space super spiky?





- At layer 14, PC1 and mean have ~0.98 cosine similarity
- Yet, perturbing along mean looks random, perturbing along PC1 looks like feature!!

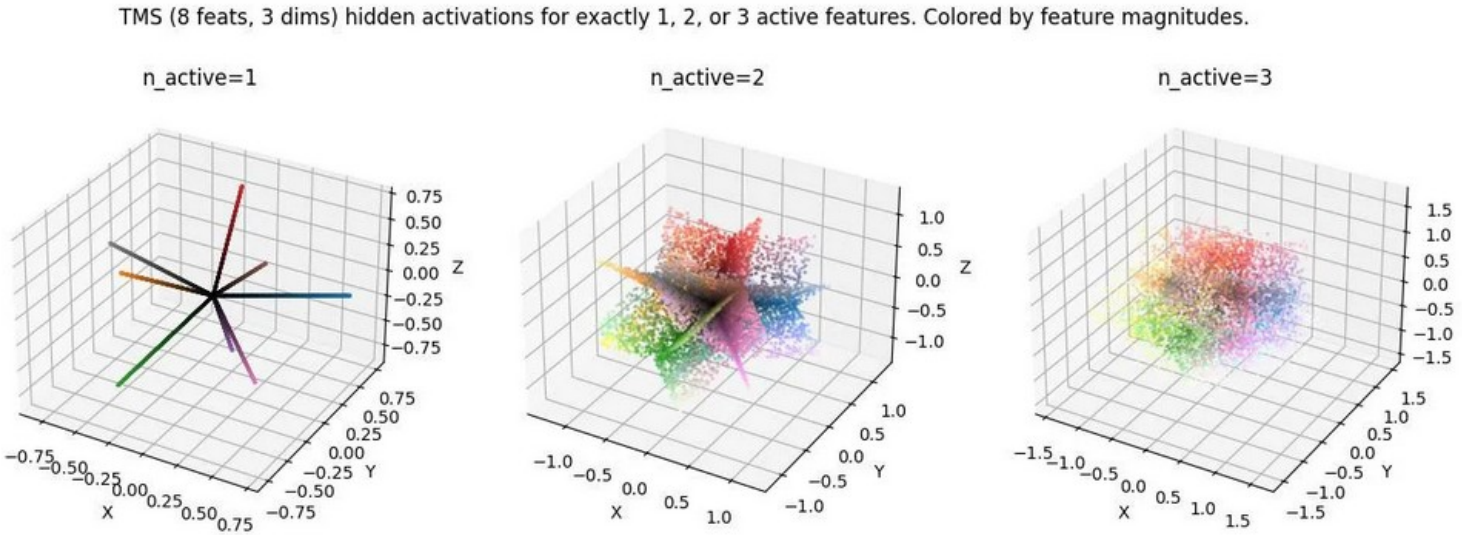
Steering behavior of PC1, MELBO_1, Mean

- Surprisingly, not that different?
- Steering along these directions produces the expected change - model switches between English and Mandarin
- slightly easier (less magnitude required) with MELBO_1, but not a big difference
- Preliminary and qualitative; needs more work

Strikes me as unexpected? I would expect steering along the mean direction to require much more magnitude to work?

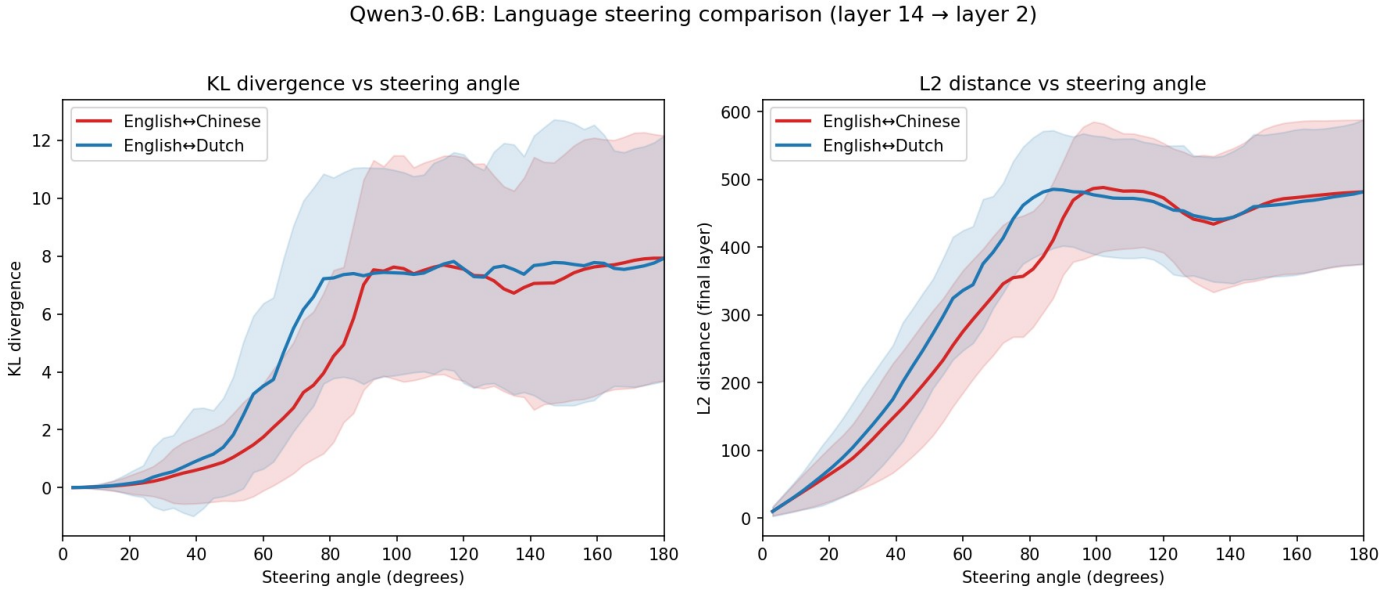
Is activation space super spiky? People think so!

Stolen from Stefan's LessWrong
These results support this picture, I think? And that PC1/MELBO_1 is a feature?



Short aside: similar behavior with different languages

I was worried that steering looking random had to do with English and Mandarin being very different -> hence try closer languages

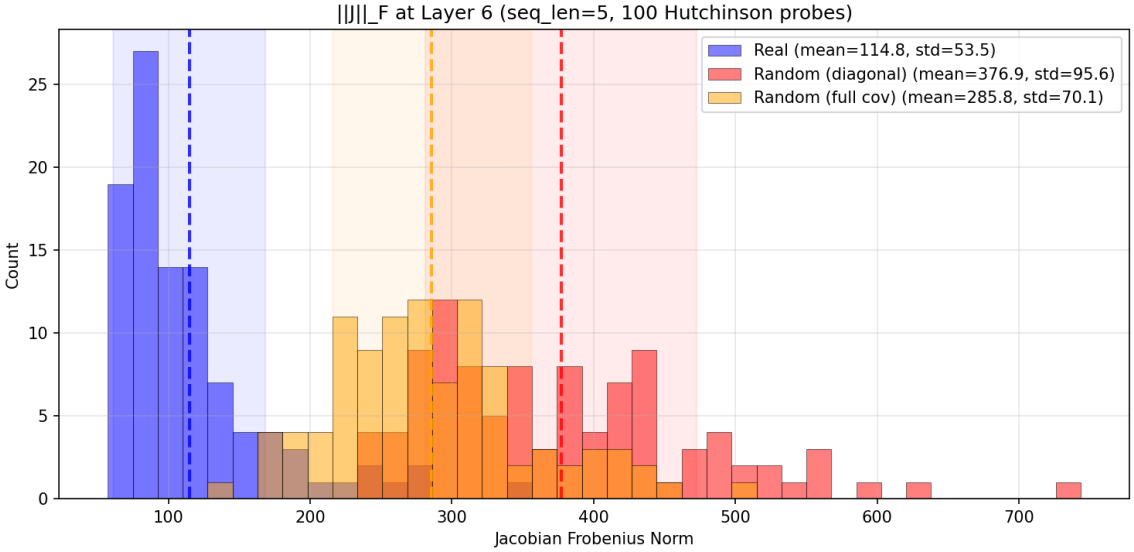


Looks similar? Dutch does respond 'more' but idk this is not an in-depth investigation

Questions and directions

- Steering is autoregressive: activations change at each step, so the exact feature direction shifts
- Hypothesis: Fixed directions (MELBO_1, PC1, mean) all point roughly right, so they average out and steer similarly despite very different single-shot perturbation behavior
- Testable: does MELBO_1 still break the plateau on held-out activations? If yes, it's a globally special direction. If no, the single-shot result reflects activation-specific alignment, not a cleaner feature
- If not globally special, testable prediction: for same 'steering magnitude budget', many small pushes (e.g., steer at every token generation) work better than few big pushes (e.g., steer 5x as strong every 5th token)
- Quick test: steer with activation shift vector (first slide); does it do something interpretable?

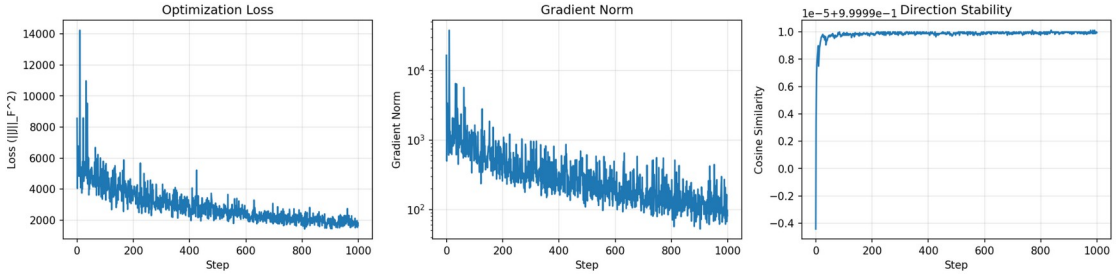
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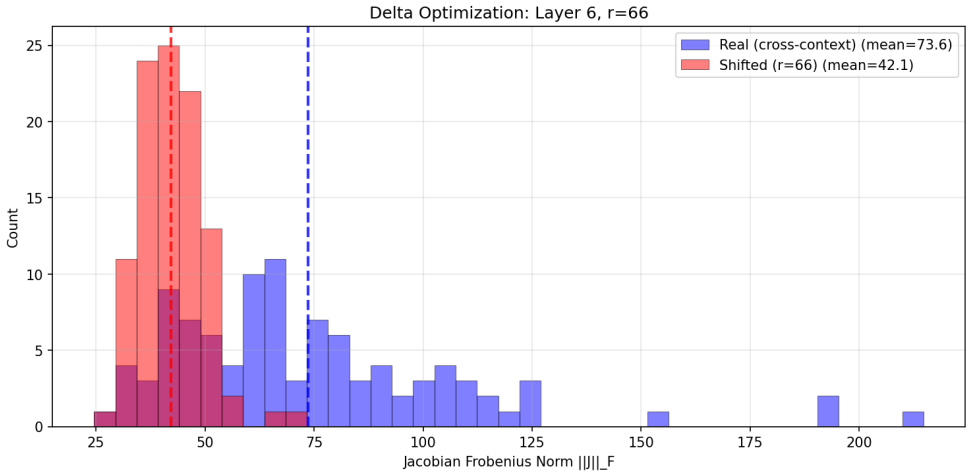


Last time: Matching first two moments of Gaussian does not result in real-like activations

Up next: Can we optimize for vector that takes us from set of real activations to set of real-like activations, as measured by Jacobian norm distribution?

Yes! With a caveat…





Method:

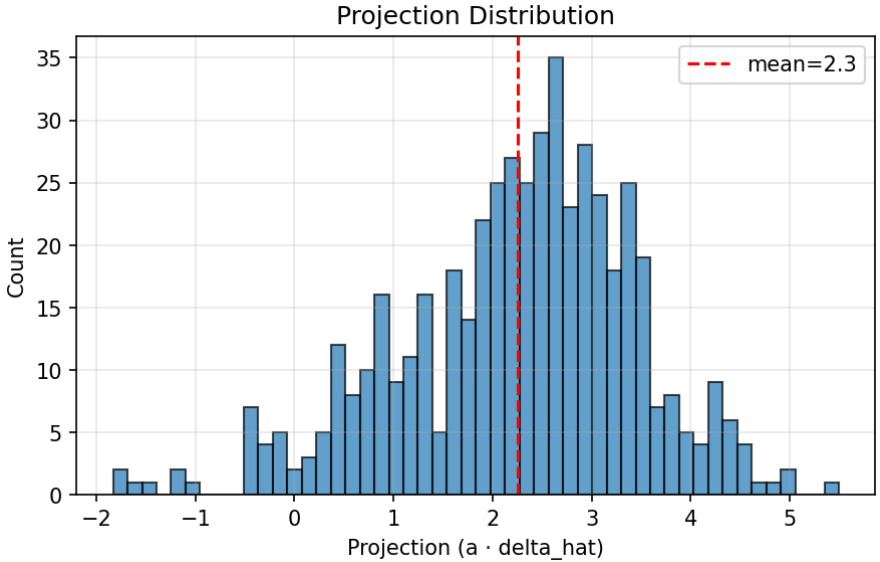
- Minimizing Jacobian of last layer activations w.r.t. Layer 6 activations, last-token only
- Fix magnitude to mean inter-activation magnitude; optimize direction

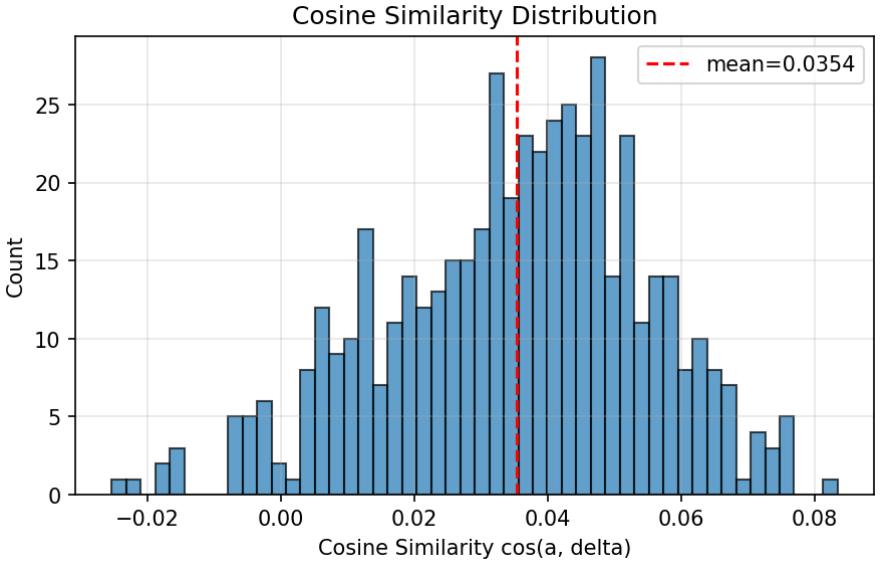
Spot the mistake? :-) Wrong optimization target: we want real-like, _not_ smallest Jacobian

Std 34 -> 7

Does the resulting vector mean anything?

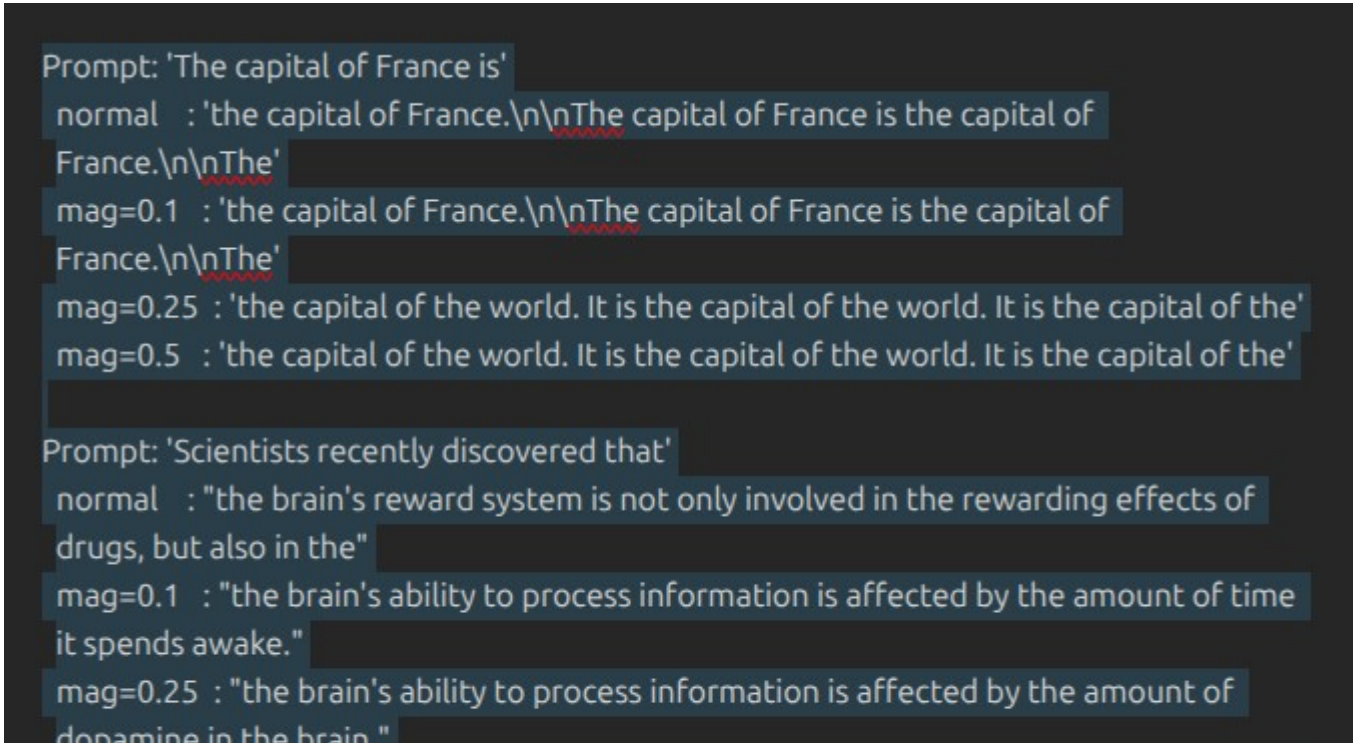
I'm not sure, but I don't think so. Are these even the right metrics? (This is over 100 real activations)





Does the resulting vector mean anything?

I'm not sure, but I don't think so. Some primitive steering:



Next steps and questions

- Obvious next step: do the optimization with the right target :-)
  - Potential 2nd next step: match also std of Jacobian norm distribution
  - Potential nth next steps: optimize for other properties (directional?)
- Q: Thoughts on methodology? Valuable to sweep for different values of magnitude?
  - Trivial for small (stay in plateau); perhaps impossible at some point? Idk SGD strong
- Q: Is 'does this (resulting) vector mean anything the right question'? What about 'what are these activations that we find?'?
- Q: How to answer the above questions?
  - Projections onto real activations?
  - Steering-like experiments and look at output?
  - Other things I haven't thought of?